Dhanushvarman J

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SUMMARY

A highly skilled and results driven **Frontend Developer** with **3+ years** of experience specializing in the React ecosystem. Proven ability to build and maintain scalable applications, with expertise in implementing advanced features such as Progressive Web Apps (PWAs) with offline support, real-time data visualization, and automated PDF reporting. Adept at leading projects and collaborating directly with clients to deliver high-quality, impactful user experiences.

SKILLS

Frontend Development: React JS, Redux, Redux-Saga, RTK Query, React Native, Ant Design, Bootstrap, Tailwind CSS, Formik
Programming & Web: JavaScript (ES6+), Typescript, HTML5, CSS3 and Node.js
API Integration: Reusable REST service layers with axios, structured error handling
Tools & Platforms: Git, npm, VS Code, Chrome DevTools
Version Control & Collaboration: Git/GitHub branching workflows, feature branches, code reviews
Languages: Tamil (Native), English (Fluent), Hindi (Basic/Understandable)

WORK EXPERIENCE

Frontend Developer (UI/UX) — Chennai, India
Mandark Technologies Private Limited *(website)* — *April 2023 - Present*

- Built and maintained **scalable React JS applications** using **Redux/Redux-Saga, Ant Design, Tailwind, and Bootstrap.**
- Integrated **RTK Query** across core frontend applications to streamline state management, caching, and cross-component data synchronization.
- Developed apps with **React Native**, integrating **Google Maps** for location-based features and seamless mobile experience.
- Implemented **web-based map interactions** using **Leaflet** for dynamic and real-time data visualization.
- Implemented **Progressive Web App (PWA)** with robust **offline capabilities** for tablet users, leveraging **Service worker** and **IndexedDB** for caching data without internet connection.
- Developed an **advanced error capture component** using **rrweb** to record DOM events (with a 1 minute circular buffer), **html2canvas** for page screenshots, and **Axios** interceptors for capturing console logs and API calls (including responses) during errors.
- Integrated **React Flow** to create interactive diagrams for visualizing energy flow and component connections.
- Developed **dynamic data visualizations** with **Plotly.js**, enabling real-time charting and advanced analytics for large datasets.
- Automated **PDF reports** generation using the **Puppeteer** library, incorporating page breaks, page numbers, and dynamic content logging on each page for download.
- Conducted knowledge-sharing sessions on Puppeteer based document services and Clean Code practices to improve development standards.
- Integrated secure authentication and authorization workflows using **JWT tokens** and role-based access.
- Collaborated directly with clients to gather requirements, clarify use cases, and provide technical inputs during feature planning.
- **Planned and tracked sprints** using GitHub Projects/Sprint Boards, ensuring timely delivery of features and bug fixes.

Project Associate — Chennai, India
Indian Institute of Technology Madras (IITM) — *October 2025 - March 2026*

- Worked under the guidance of **Dr. Nikhil Bugalia** on a logistics optimization initiative for city-wide solid waste management.
- Developed and deployed an interactive web-based simulation platform to address the **Vehicle Routing Problem (VRP)** for dynamic waste collection across Chennai.
- Engineered features that allowed users to input diverse datasets, run complex routing simulations, and visually track optimized vehicle paths.

MAJOR PROJECTS

IEAC (Industrial Energy Assessment Cell) — *Team Lead & Front-End Developer* *April 2023 – Present*

- Led the project team, managed client communications, and planned feature roadmaps.
- Engineered and deployed the **IEAC Audit Mobile App** using **React Native**, tailored specifically for field auditors.
- Architected robust **offline capabilities and background data syncing** for the mobile app, utilizing **Drizzle ORM (SQLite)** for highly efficient local relational data storage and integrity management.
- Successfully oversaw the conversion and release of Version 2, enhancing existing functionalities across both web and mobile platforms.
- Conducted training and walkthrough sessions for auditors to ensure smooth adoption of the application.

SGRI (Saint Gobain Research India) — *Front-End Developer* *August 2023 – February 2024*

- Implemented the frontend interface for a data visualization platform.
- Developed functionality to upload Excel sheets for structured data handling.
- Integrated **Plotly.js** to generate interactive and customizable charts from uploaded datasets.
- Enhanced usability by enabling real-time data exploration and analysis through dynamic visualizations.

GCTP (Greater Chennai Traffic Police) — *Project Owner & Front-End Lead* *April 2023 – Present*

- Developed and led the **website and mobile app** for traffic monitoring across Chennai in collaboration with the Police Commissioner.
- Incorporated **Leaflet Maps** to provide real-time location tracking and visualization of traffic flow and incident reports.
- Developed a mobile app for officers to log in at their assigned junctions and report incidents such as accidents or roadblocks to the control room.
- Currently working on converting from **Version 1 to Version 2** (yet to be released).

CO - CURRICULAR ACTIVITIES

IEAC Audit Tool — Product Showcase & User Engagement — *Annual Energy Summit* *December 2024*

- Showcased the IEAC product at **IIT Gandhinagar** during the KISEM Energy Meet.
- Gave a demo of the application at the **Annual Energy Meet held in IIT Madras** to industry professionals and auditors.
- Engaged directly with auditors in the field to understand their challenges and demonstrate how the tool streamlines their audit process.

GCTP — Problem Solving & Implementation

- Gained insights into the challenges faced by **Chennai Traffic Police** through direct meetings with control room officers.
- Coordinated with officers to deliver a tool enabling them to analyze traffic across Chennai and optimize route planning.
- Presented a comprehensive demo of the application to the entire **Chennai Traffic Police officers at the Commissioner's office.**

EDUCATION

Sri Sivasubramaniya Nadar College of Engineering (SSN) CGPA : 7.9 / 10
Bachelor of Technology in Civil Engineering *July 2018 - May 2022*